

A CONFIRMATORY AUDIT

4 public endpoints

BBBP | ClinTox
ESOL | Lipophilicity

3 split designs

Random | scaffold
similarity cluster

5 reliability lenses

Performance | calibration
domain | sets | retention



B FAILURE SIGNALS

0.90 overall

0.07-0.10 positive
ClinTox marginal coverage

~71% ambiguous

Mondrian repairs coverage
by returning {0,1}

27-31% retained

ClinTox positives at
50% overall retention



C TAME DESIGN

Two label-free views

Physicochemical
descriptors + scores

Transport certificate

Support | ESS | clipping
AUC | weighted balance

Protected envelope

Consensus expansion
exact fallback on failure



D SEALED EPA EVALUATION

Prediction firewall

HASH + SEAL
then open labels

+1.36 pp

Minimum-class coverage
95% interval:
+0.58 to +2.01 pp

MacroCSY -1.61 pp

Inside frozen -5 pp
efficiency guardrail

60/60 final endpoint-seed cells completed under the prediction-to-label firewall